

AEGIS-AI

Detect Cyber Threats Before They Reach You

Team	Yashika H · Mohith BN
Built for	IBM Bobathon / Hackathon — Cybersecurity Track
Powered by	IBM Bob (Enterprise LLM)

Drop any file, screenshot, or image. AEGIS-AI extracts text with OCR, runs it through IBM Bob's language model, and returns an instant risk level — in plain English, with zero jargon.

Features

- **AI-powered threat detection** — upload a screenshot, image, PDF, or paste a URL/text for instant analysis
- **OCR text extraction** — Tesseract-based pipeline reads every character from a file, even low-resolution or tilted images
- **IBM Bob semantic analysis** — an enterprise-grade LLM classifies threat category and intent at the semantic level, not just keyword matching (understands intent, tone, urgency, and contextual red flags)
- **Five-tier risk scoring** — every result is mapped to a clear, colour-coded risk level: Safe, Low, Medium, High, or Critical
- **Plain-language verdicts** — every result includes a threat category, a jargon-free explanation, and concrete safety recommendations
- **Live Scanner** — a real-time pipeline (Upload → OCR → IBM Bob analysis → Risk score) that returns a verdict in under 3 seconds
- **Threat Library** — a searchable, filterable reference guide covering today's top threats: Phishing, Business Email Compromise, Smishing, QR Code Quishing, Malware Links, Credential Harvesting, and Data Exfiltration Indicators
- **Privacy by default** — files are processed in-memory and immediately discarded; nothing is stored, logged, or shared, and no account is required

Tech Stack

Frontend

- React
- Vite
- Framer Motion — smooth, motion-safe scroll animations

Backend

- Python
- Flask — lightweight REST API, no data persistence

AI / Intelligence

- IBM Bob — enterprise language model used for threat classification, intent and tone analysis
- Tesseract / PIL — OCR engine for pixel-to-text extraction from any image format

How to Run Locally

Frontend

```
cd aegis-frontend  
npm install npm  
run dev
```

Backend

```
cd aegis-backend  
pip install -r requirements.txt  
python app.py
```

The frontend runs on Vite's local dev server (typically `localhost:5173`) and communicates with the Flask REST API on the backend. Update the API base URL in the frontend environment config if your backend runs on a different port.

Problem

Cyber threats don't discriminate by technical knowledge — but the tools to detect them do. Phishing, Business Email Compromise, smishing, and QR-code quishing attacks are engineered to look convincing, and everyday users are rarely equipped with the security expertise or tools to tell a real message from a malicious one until it's too late.

Solution

AEGIS-AI bridges the gap between sophisticated cyber threats and everyday users who lack the tools to identify them. Drop in a screenshot, file, or link and the pipeline takes over end-to-end: OCR extracts every readable character, IBM Bob's enterprise LLM analyses intent, tone, and contextual red flags, and the result is scored into one of five clear risk tiers with a plain-language explanation and next steps — all in under 3 seconds, with nothing stored. AEGIS-AI gives everyone the same defensive edge, regardless of technical background.

Team

Yashika H	Developer
Mohith BN	Developer

(Update roles above as needed.)

Future Scope

- Browser extension for real-time link and page scanning
- Mobile app companion for on-the-go screenshot scanning
- Multi-language OCR and threat detection support
- Email client plugin (Gmail / Outlook) for inline inbox scanning
- Community-reported threat database to crowd-source new attack patterns
- Expanded Threat Library with live threat-intelligence feed integration

Disclaimer: AEGIS-AI is an educational and awareness tool. Results are AI-generated and should be treated as guidance, not definitive legal or forensic analysis. For confirmed security incidents, contact your organisation's security team or a qualified professional.

No jargon. No accounts. No data retention.

AEGIS-AI: cybersecurity awareness, powered by AI.